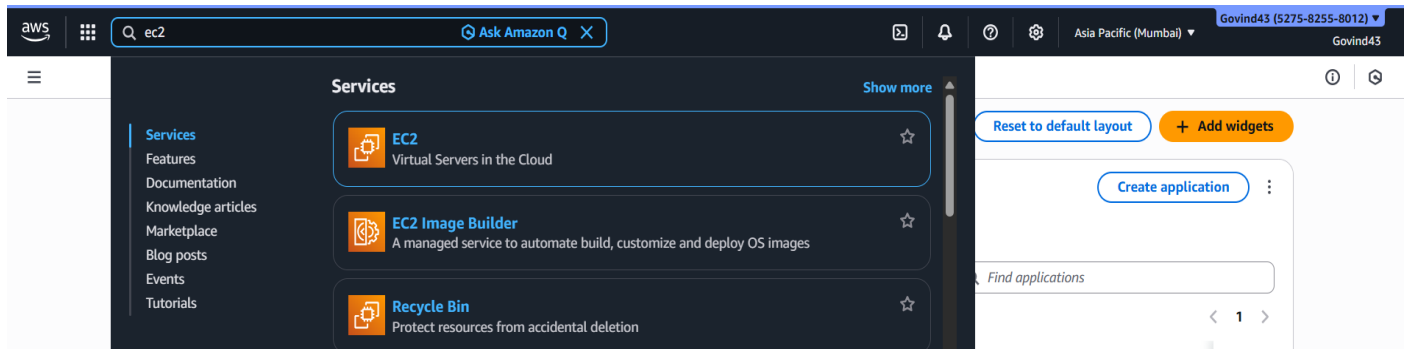


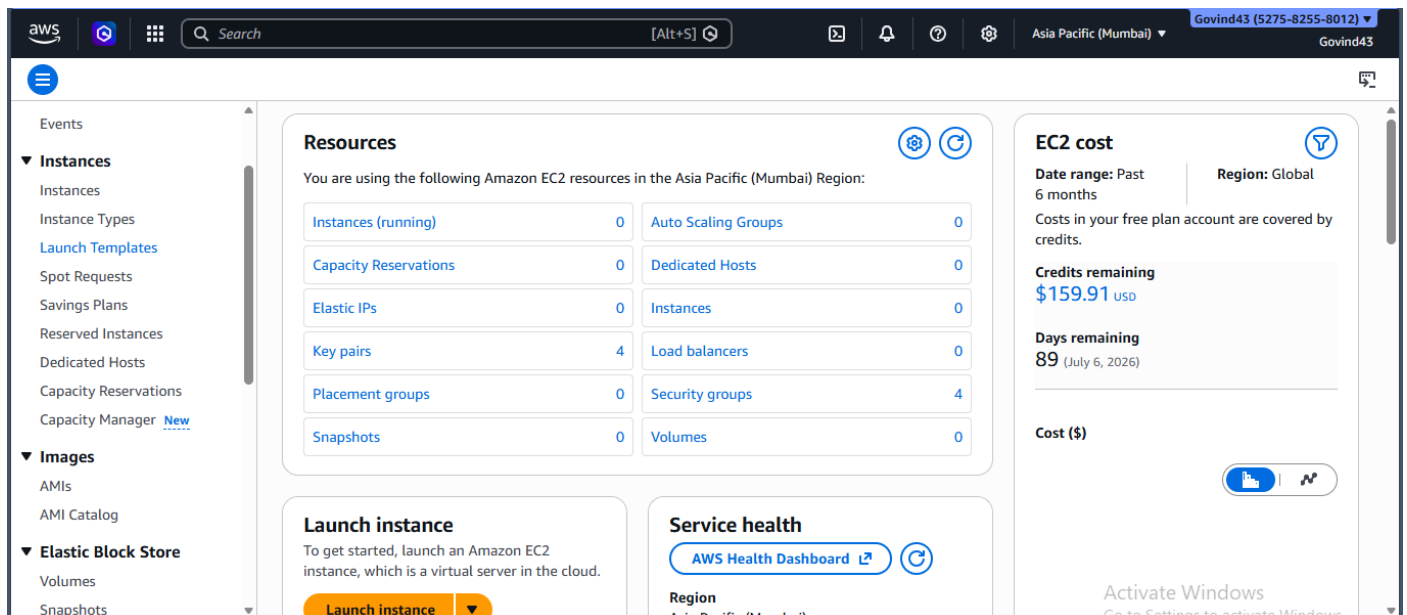
11. PROBLEM STATEMENT:

Build scaling plans in AWS that balance load on different EC2 instances.

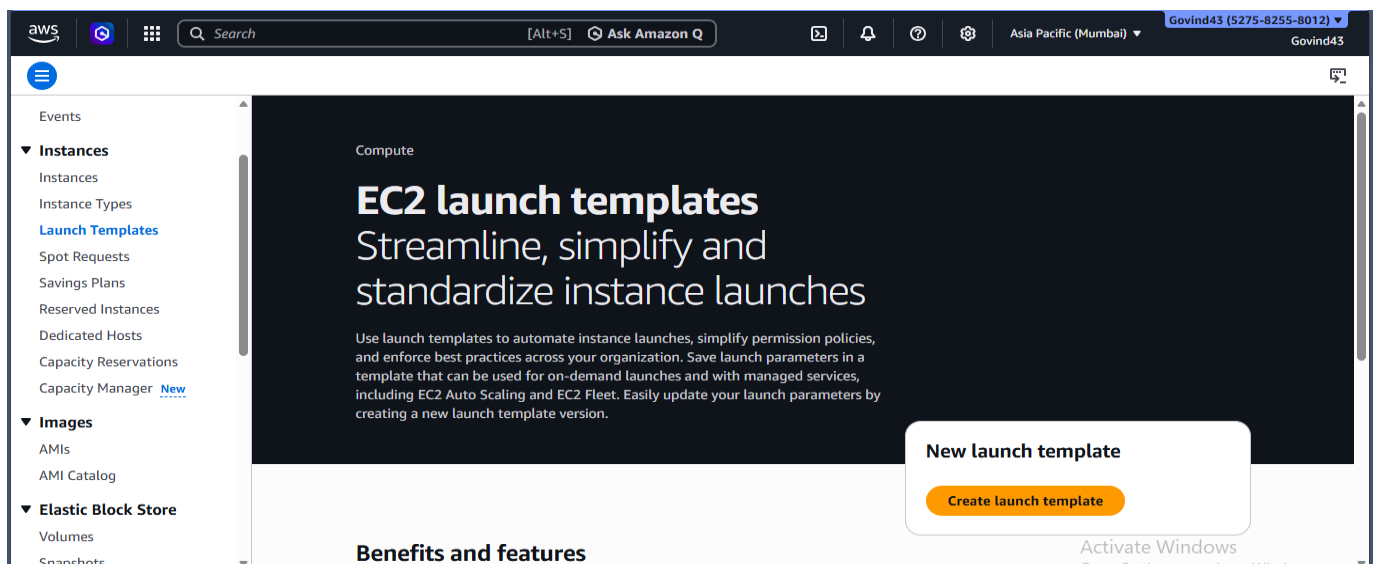
1. Log into AWS console and click on “EC2”.



2. Click on ‘Launch Templates’.



3. Create launch templates.



4. Enter necessary details. In 'Launch templet name and description' give a name and description to your launch templet and select the check box for 'Provide guidance to help me set up a template that I can use with EC2 Auto Scaling'

Create launch template

Creating a launch template allows you to create a saved instance configuration that can be reused, shared and launched at a later time. Templates can have multiple versions.

Launch template name and description

Launch template name - *required*

template1

Must be unique to this account. Max 128 chars. No spaces or special characters like '&', '*', '@'.

Template version description

template1

Max 255 chars

Auto Scaling guidance | [Info](#)

Select this if you intend to use this template with EC2 Auto Scaling

☒ Provide guidance to help me set up a template that I can use with EC2 Auto Scaling

Template tags

Summary

Software Image (AMI)

-

Virtual server type (instance type)

-

Firewall (security group)

-

Storage (volumes)

-

[Cancel](#) [Create launch template](#)

For OS Image select 'Ubuntu Server'

Create launch template

Search our full catalog including 1000s of application and OS images

Quick Start

Amazon Linux macOS **Ubuntu** Windows Red Hat SUSE Linux Debian

[Browse more AMIs](#)

Including AMIs from AWS, Marketplace and the Community

Summary

Software Image (AMI)

Canonical, Ubuntu, 24.04, amd64...[read more](#)

ami-05d2d839d4f73aafb

Virtual server type (instance type)

-

Firewall (security group)

-

Storage (volumes)

-

For instance type, select 't2.micro' or any other free tier eligible instance type.
For key pair, select an existing key pair or crate a new key pair.

Create launch template

Instance type | [Info](#) | [Get advice](#) [Advanced](#)

Instance type

t3.micro

Family: t3 2 vCPU 1 GiB Memory Current generation: true

On-Demand Linux base pricing: 0.0112 USD per Hour On-Demand SUSE base pricing: 0.0112 USD per Hour

On-Demand Windows base pricing: 0.0204 USD per Hour

On-Demand Ubuntu Pro base pricing: 0.0147 USD per Hour

On-Demand RHEL base pricing: 0.04 USD per Hour

Free tier eligible

☐ All generations

[Compare instance types](#)

[Additional costs apply for AMIs with pre-installed software](#)

Key pair (login) | [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name

gbs.keypair

☐ Create new key pair

Summary

Software Image (AMI)

Canonical, Ubuntu, 24.04, amd64...[read more](#)

ami-05d2d839d4f73aafb

Virtual server type (instance type)

t3.micro

Firewall (security group)

-

Storage (volumes)

1 volume(s) - 8 GiB

[Cancel](#) [Create launch template](#)

For security group select an existing security group where port 4000 is open for incoming traffic.

Keep Storage and Resource Tag as default.

The screenshot shows the 'Create launch template' page in the AWS Management Console. The 'Network settings' section is expanded, showing options for Subnet, Availability Zone, Firewall (security groups), and Security groups. The 'Firewall (security groups)' section is selected, and 'security2' is chosen from the dropdown. The 'Summary' panel on the right shows the configuration: Software Image (AMI) as Canonical, Ubuntu, 24.04, amd64; Virtual server type (instance type) as t3.micro; Firewall (security group) as security2; and Storage (volumes) as 1 volume(s) - 8 GiB. The 'Create launch template' button is visible.

In Advanced details scroll down to the bottom and add the text to the user data. Then click on Create launch template.

The screenshot shows the 'Create launch template' page with the 'User data - optional' section expanded. The 'Choose file' button is visible, and a text area contains the following script:

```
#!/bin/bash
apt-get update
apt-get install -y nginx
systemctl start nginx
systemctl enable nginx
apt-get install -y git
curl -sL https://deb.nodesource.com/setup_16.x | sudo -E bash -
apt-get install -y nodejs
git clone https://github.com/itsgbs/cse1project.git
cd cse1project
npm install
node index.js
```

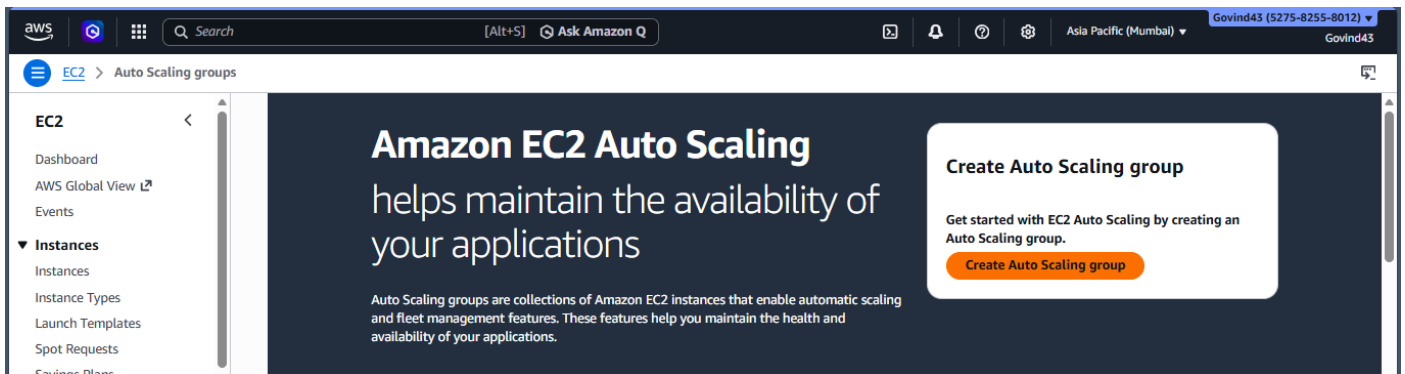
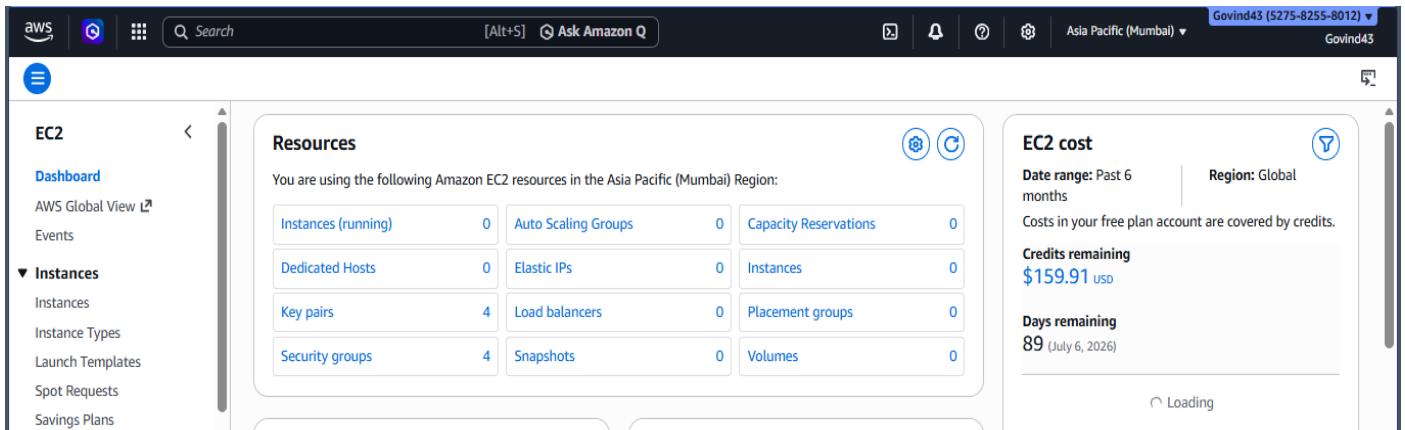
The 'Summary' panel on the right shows the same configuration as the previous screenshot. The 'Create launch template' button is visible.

Launch Templates created successfully.

The screenshot shows the 'Launch Templates' page in the AWS Management Console. A table lists the created launch templates.

Launch Template ID	Launch Template Name	Default Version	Latest Version	Create Time	Created By
lt-08ccc353ee465e483	template1	1	1	2026-04-08T05:37:51.000Z	arn:aws:iam::527

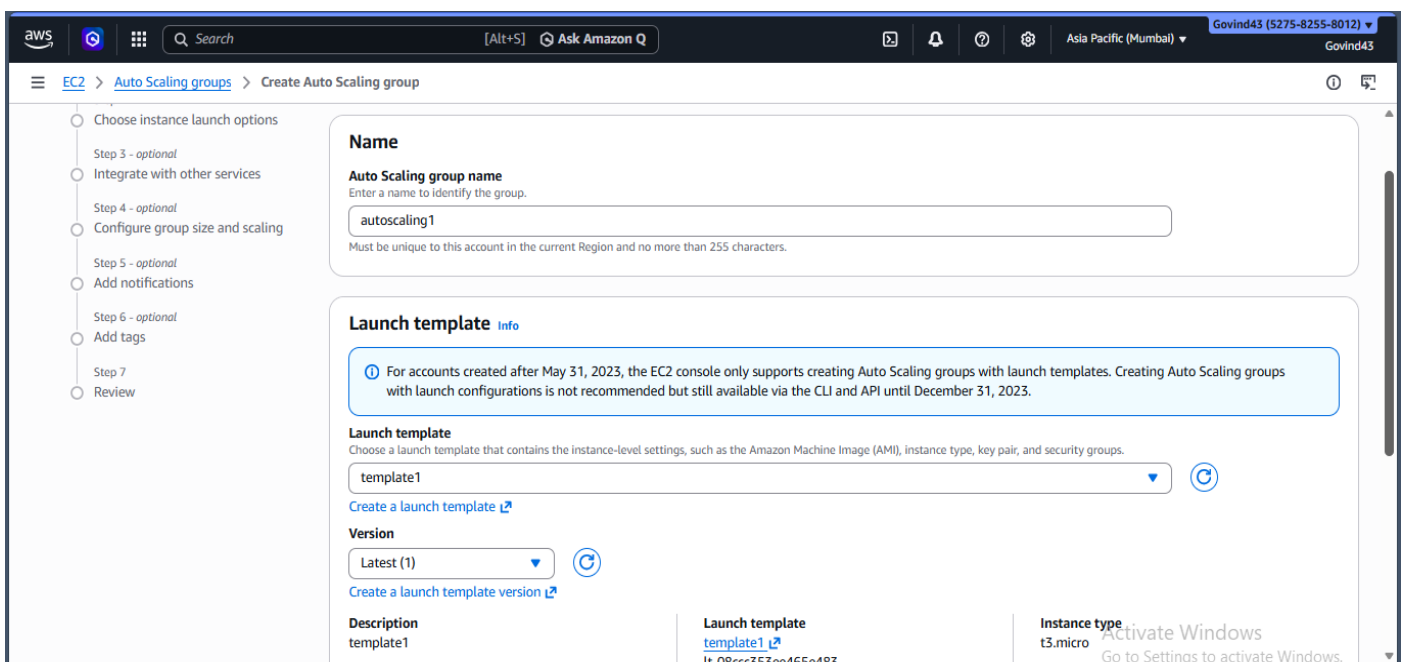
6. Now go to 'Auto Scaling Groups' . Then click on 'Create Auto Scaling Group'.



7. Add Details for the Auto Scaling Group.

Step 1: Choose launch templates- Give an name to the Auto Scaling Group and choose the Launch template you just created as the launch template.

Click on Next.



Step 2: Choose instance launch options- Choose all the 'Availability Zones' available. Click on Next.

The screenshot shows the 'Create Auto Scaling group' page in the AWS Management Console, specifically the 'Choose instance launch options' step. The page is for the 'Govind43' account in the 'Asia Pacific (Mumbai)' region. The 'VPC' section shows 'vpc-0811d035c1c21c10d' selected. The 'Availability Zones and subnets' section shows three zones selected: 'aps1-az1 (ap-south-1a)', 'aps1-az2 (ap-south-1c)', and 'aps1-az3 (ap-south-1b)'. The 'Availability Zone distribution' section shows 'Balanced best effort' selected. The 'Load balancing' section shows 'Attach to a new load balancer' selected. The 'Load balancer type' section shows 'Application Load Balancer' selected. The 'Load balancer name' field contains 'autoscaling1-1'. The 'Load balancer scheme' section shows 'Internet-facing' selected. The 'Listeners and routing' section shows 'HTTP' selected for the protocol and '80' for the port. The 'Default routing (forward to)' section shows 'Create a target group' selected. The 'New target group name' field contains 'autoscaling1-1'. The 'Tags' section shows 'Add tag' button.

Step 3: Configure advanced options- For Load balancing select 'Attach to a new load balancer'

The screenshot shows the 'Create Auto Scaling group' page in the AWS Management Console, specifically the 'Configure advanced options' step. The page is for the 'Govind43' account in the 'Asia Pacific (Mumbai)' region. The 'Load balancing' section shows 'Attach to a new load balancer' selected. The 'Load balancer type' section shows 'Application Load Balancer' selected. The 'Load balancer name' field contains 'autoscaling1-1'. The 'Load balancer scheme' section shows 'Internet-facing' selected. The 'Listeners and routing' section shows 'HTTP' selected for the protocol and '80' for the port. The 'Default routing (forward to)' section shows 'Create a target group' selected. The 'New target group name' field contains 'autoscaling1-1'. The 'Tags' section shows 'Add tag' button.

For new load balancer configuration select 'Application Load Balancer', 'Internet-facing' and make HTTP port from 80 to 4000 while creating a target group.

The screenshot shows the 'Create Auto Scaling group' page in the AWS Management Console, specifically the 'Configure advanced options' step. The page is for the 'Govind43' account in the 'Asia Pacific (Mumbai)' region. The 'Load balancing' section shows 'Attach to a new load balancer' selected. The 'Load balancer type' section shows 'Application Load Balancer' selected. The 'Load balancer name' field contains 'autoscaling1-1'. The 'Load balancer scheme' section shows 'Internet-facing' selected. The 'Listeners and routing' section shows 'HTTP' selected for the protocol and '80' for the port. The 'Default routing (forward to)' section shows 'Create a target group' selected. The 'New target group name' field contains 'autoscaling1-1'. The 'Tags' section shows 'Add tag' button.

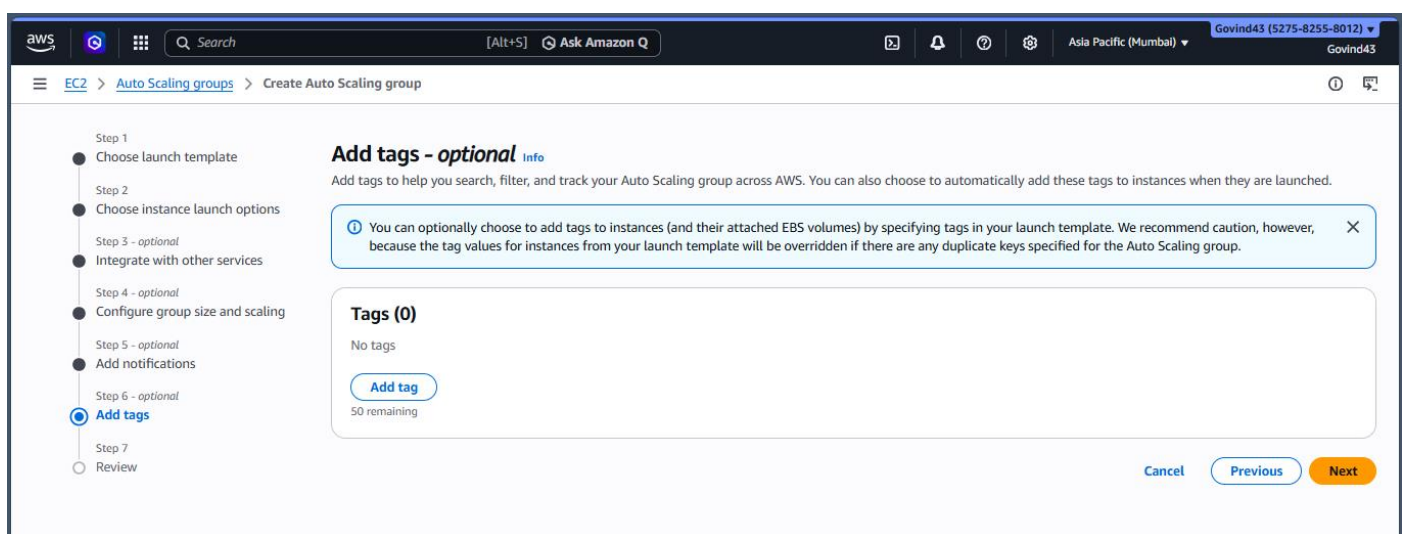
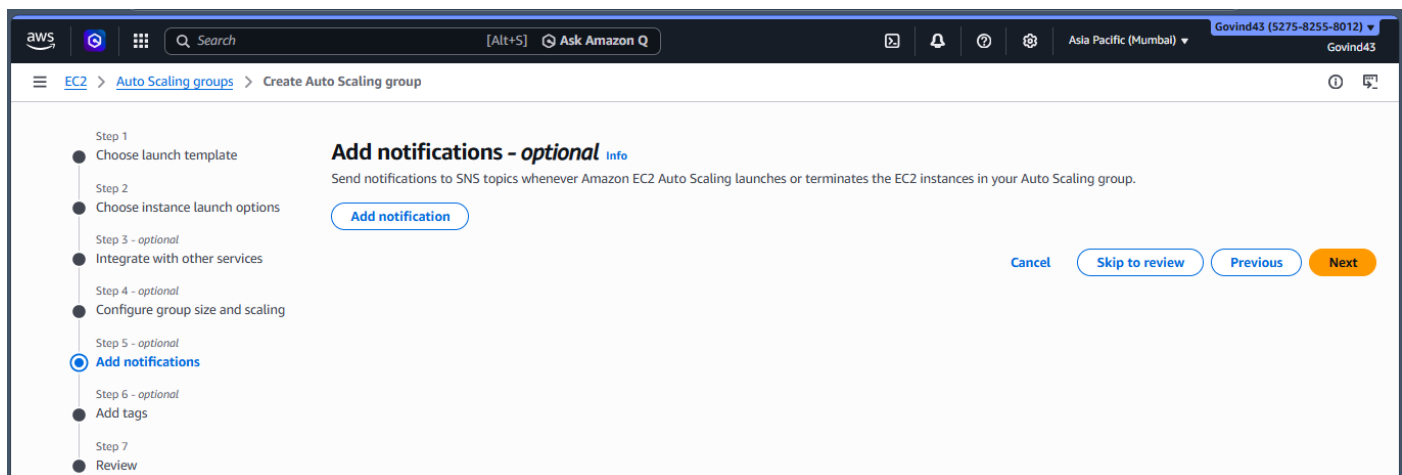
Health checks tick the check box for 'Turn on Elastic Load Balancing health checks' and make the health check grace period 180 seconds. Click on Next.

The screenshot shows the 'Health checks' section of the 'Create Auto Scaling group' wizard. It includes a description of health checks, a section for 'EC2 health checks' with a link to 'Always enabled', and a section for 'Additional health check types - optional'. The 'Turn on Elastic Load Balancing health checks' checkbox is checked and marked as 'Recommended'. A blue information box states: 'EC2 Auto Scaling will start to detect and act on health checks performed by Elastic Load Balancing. To avoid unexpected terminations, first verify the settings of these health checks in the Load Balancer console'. Other options like 'Turn on VPC Lattice health checks' and 'Turn on Amazon EBS health checks' are unchecked. The 'Health check grace period' is set to 180 seconds. At the bottom, there are 'Cancel', 'Skip to review', 'Previous', and 'Next' buttons. An 'Activate Windows' watermark is visible at the bottom right.

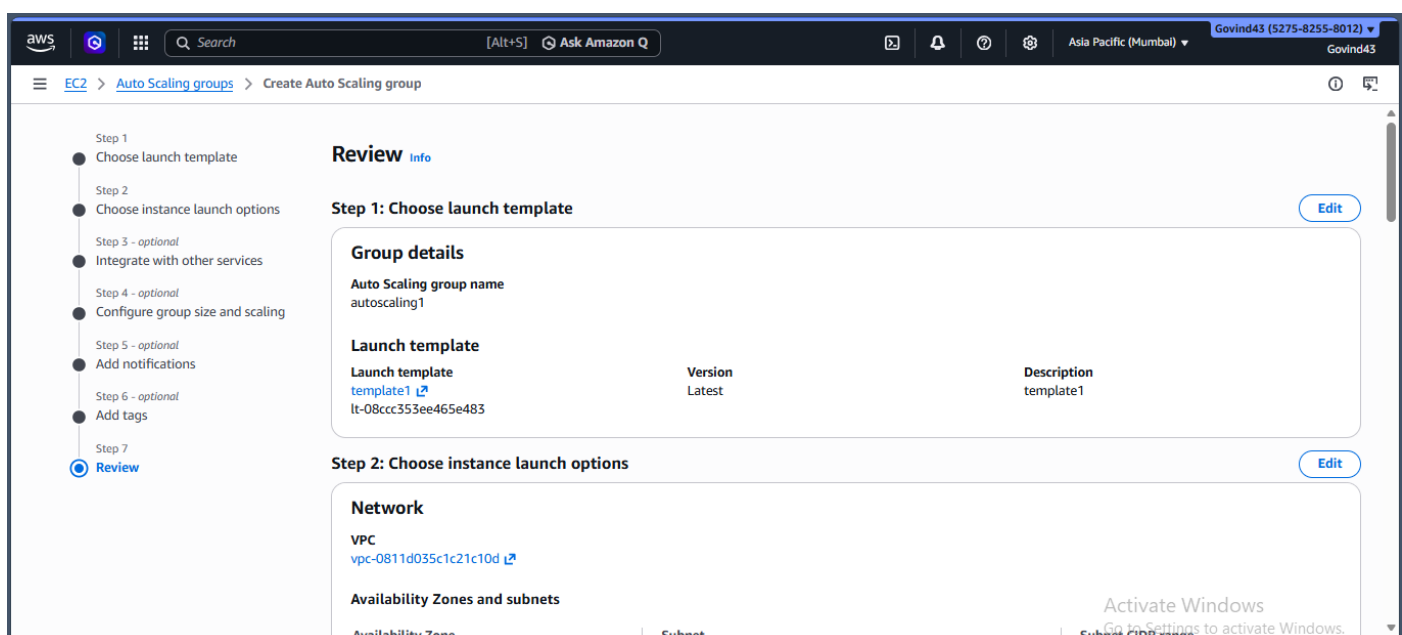
Step 4: Configure group size and scaling- In Group size set Desired capacity as 2. In Scaling set Min desired capacity as 2 and Max desired capacity as 3. In Automatic scaling choose target scaling policy and set the instance warmup as 180 second. Keeping everything else as default, Click on Next

The screenshot shows the 'Group size' and 'Scaling' sections of the 'Create Auto Scaling group' wizard. The 'Group size' section has 'Desired capacity type' set to 'Units (number of instances)' and 'Desired capacity' set to 2. The 'Scaling' section has 'Scaling limits' with 'Min desired capacity' set to 2 and 'Max desired capacity' set to 3. The 'Automatic scaling - optional' section has 'Choose whether to use a target tracking policy' set to 'Target tracking scaling policy'. The 'Scaling policy name' is 'Target Tracking Policy', the 'Metric type' is 'Average CPU utilization', and the 'Target value' is 20. The 'Instance warmup' is set to 180 seconds. There is an unchecked checkbox for 'Disable scale in to create only a scale-out policy'. An 'Activate Windows' watermark is visible at the bottom right.

Skip through Step 5 and Step 6 by clicking Next.



Finally in Step 7 Click on Create Auto Scaling Group.



Auto Scaling Group is created.

Auto Scaling groups (1) Info

Last updated less than a minute ago

Launch configurations Launch templates Actions Create Auto Scaling group

Search your Auto Scaling groups

Name	Launch template/configuration	Instances	Status	Desired capacity	Min	Max	Availability Zones
autoscaling1	template1 Version Latest	0	Updating capacity...	2	2	3	3 Availability Zones

8. Going back to instances we can see two unnamed instances are created.

Instances (2) Info

Last updated less than a minute ago

Connect Instance state Actions Launch instances

Saved filter sets: Choose filter set

Find Instance by attribute or tag (case-sensitive)

Instance state = running Clear filters

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public
	i-0b977c10770124b74	Running	t3.micro	Initializing	View alarms +	ap-south-1b	ec2-43-
	i-0548f5857486b299b	Running	t3.micro	Initializing	View alarms +	ap-south-1a	ec2-3-1

9. Copy the public IPv4 address of any one of the two instance.

Instance details | EC2 | ap-south-1

Instance summary for i-0b977c10770124b74 Info

Updated less than a minute ago

Connect Instance state Actions

Instance ID: i-0b977c10770124b74

Public IPv4 address: 43.204.227.172 | open address

Private IPv4 addresses: 172.31.3.116

Instance state: Running

Public DNS: ec2-43-204-227-172.ap-south-1.compute.amazonaws.com | open address

IPv6 address: -

Private IP DNS name (IPv4 only): ip-172-31-3-116.ap-south-1.compute.internal

Instance type: t3.micro

VPC ID: vpc-0811d035c1c21c10d

Subnet ID: subnet-061588dac00444d0e

Instance ARN: -

Hostname type: IP name: ip-172-31-3-116.ap-south-1.compute.internal

Answer private resource DNS name: -

Auto-assigned IP address: 43.204.227.172 [Public IP]

IAM role: -

IMDSv2: -

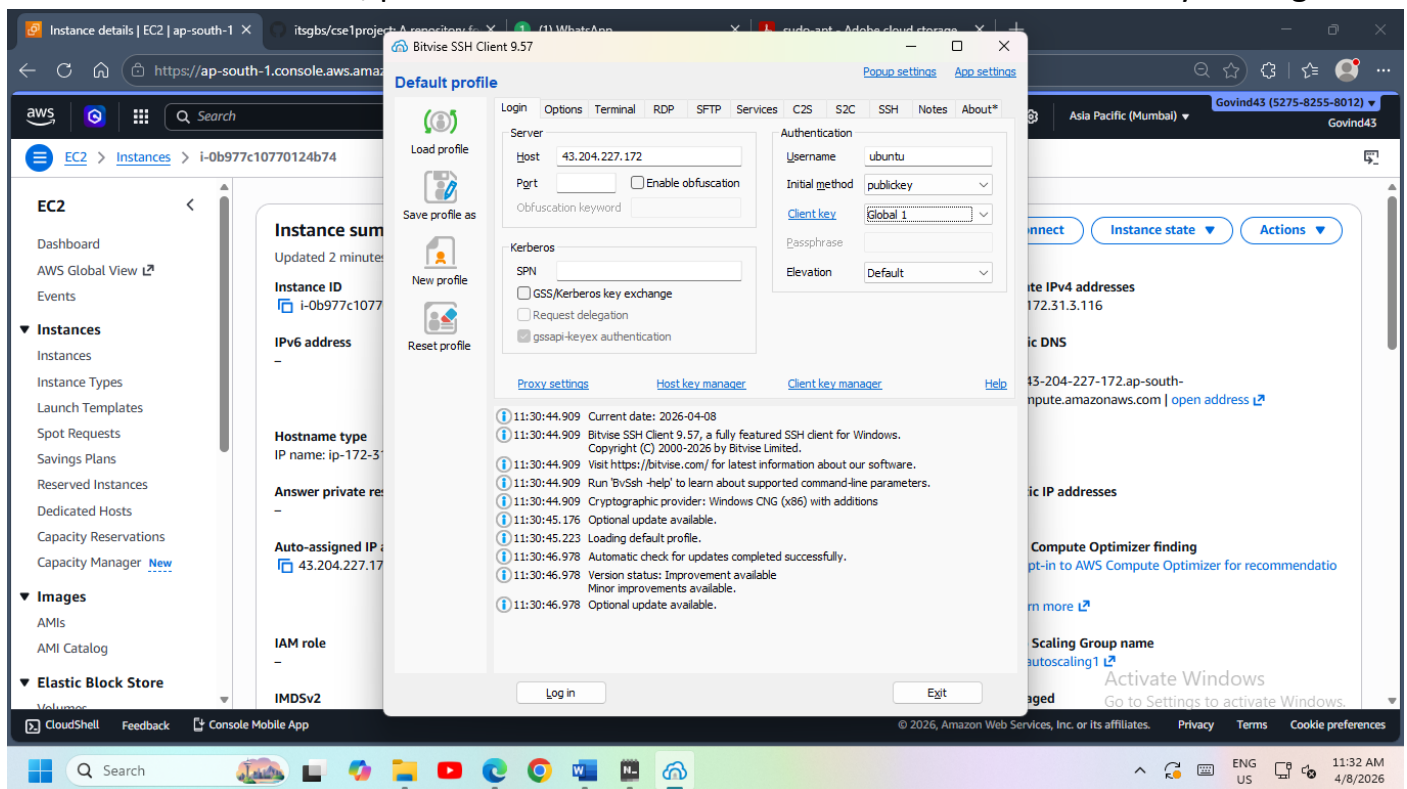
Elastic IP addresses: -

AWS Compute Optimizer finding: Opt-in to AWS Compute Optimizer for recommendations. | Learn more

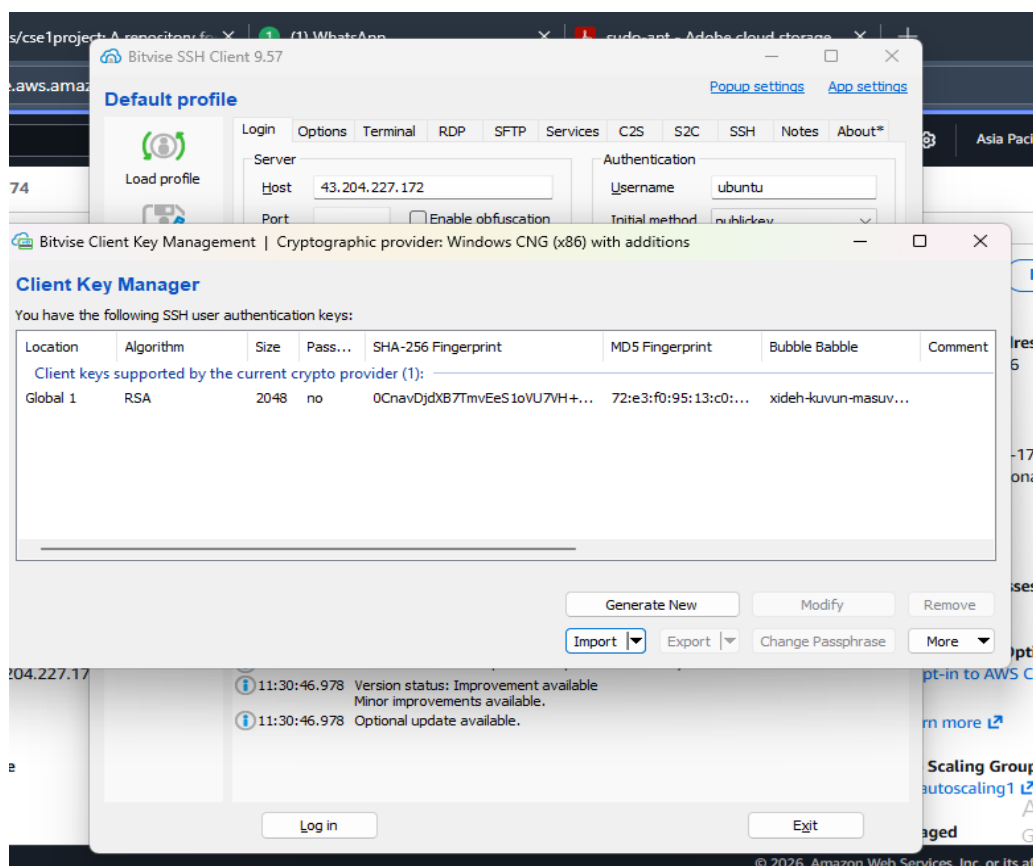
Auto Scaling Group name: autoscaling1

Managed: -

10. In Bitvise SSH Client, paste the 'Public IPV4 address' and click on 'Client key Manager'.

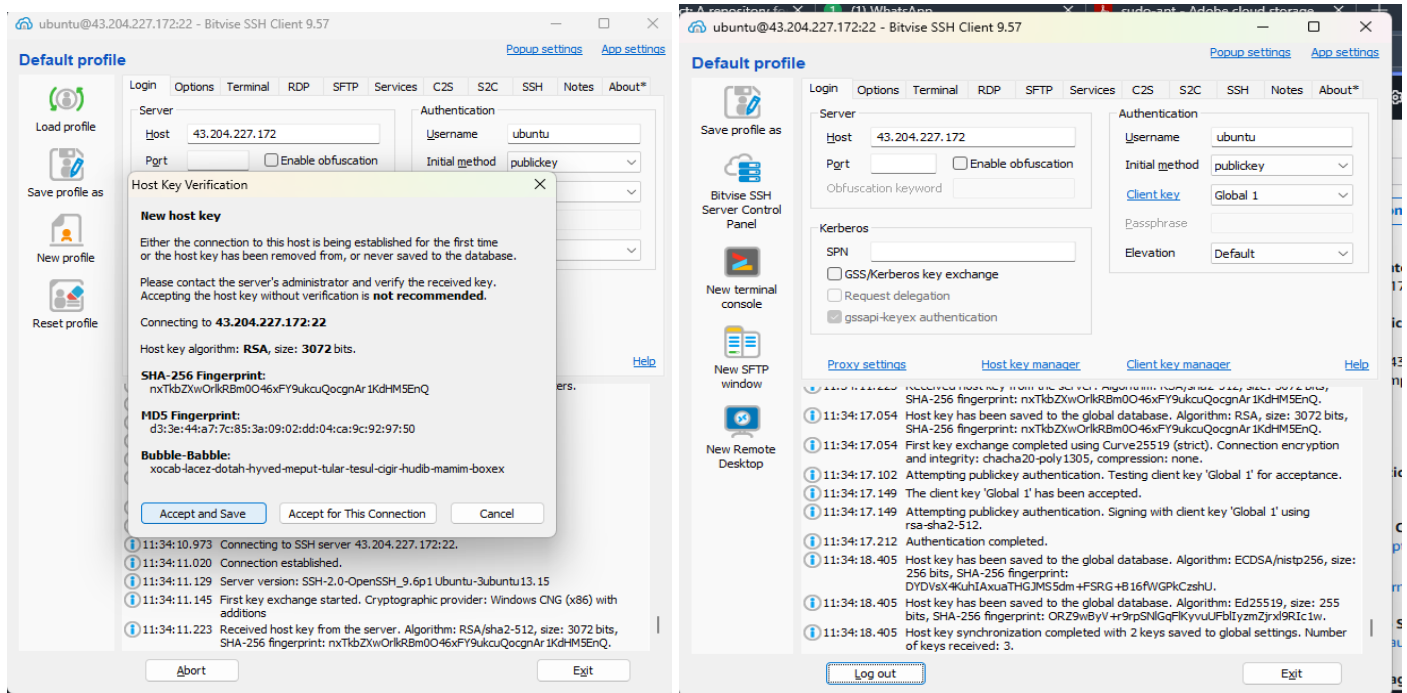


11. Under 'Client key manager', if there is any existing key then remove it and click on 'Import', select the same key that you used while creating the instance, from your file directory and then again click on 'Import' and then the key is successfully imported.



12. In 'Bitvise SSH Client', under 'Default profile' give the 'Username' as 'ubuntu' then select 'Global1' from 'Client Key Manager' then click on 'Log in' & 'Accept and save'.

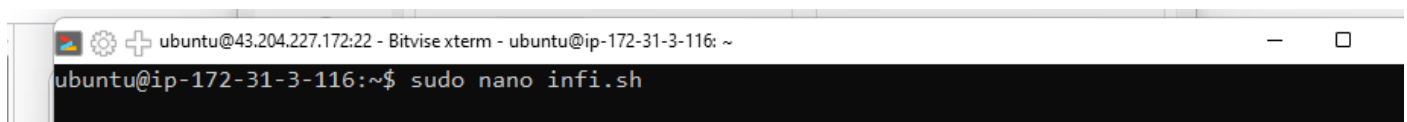
If the 'Log In' button changes to 'Log Out', then you are logged in successfully.



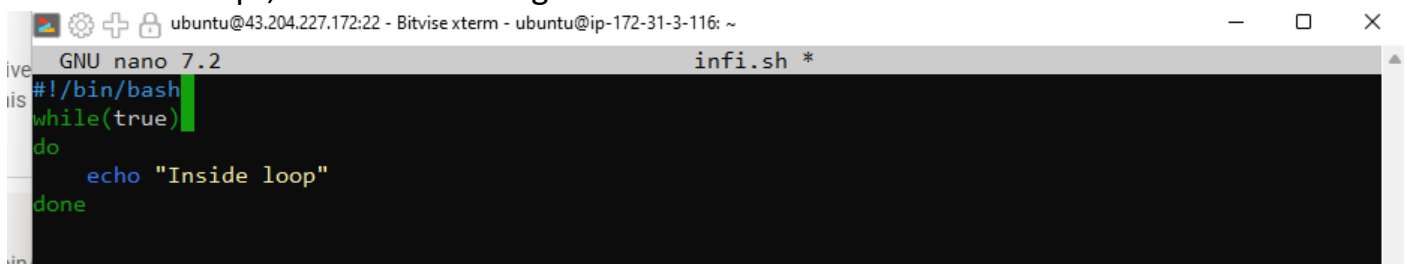
13. Click on 'New terminal consol', to access the terminal of your instance.

Once you are inside the terminal of your instance

To create a infinitely running bash script vim infi.sh

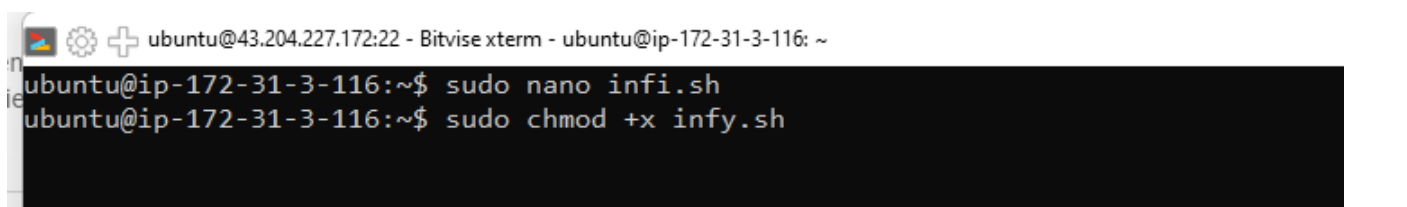


In the bash script, add the following code.



To make the script executable

sudo chmod 777 infi.sh



```
ubuntu@ip-172-31-3-116:~$ sh infi.sh
```

[illegible]

Instances | EC2 | ap-south-1

https://ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#Instances:instanceState=running

EC2 > Instances

Instances (2/2) Info

Last updated 2 minutes ago

Connect Instance state Actions Launch instances

Saved filter sets

Choose filter set Find Instance by attribute or tag (case-sensitive)

Instance state = running Clear filters

☒	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public
☒		i-038c22fc550ffc76	Running	t3.micro	3/3 checks passed	View alarms +	ap-south-1b	ec2-15-
☒		i-0548f5857486b299b	Running	t3.micro	3/3 checks passed	View alarms +	ap-south-1a	ec2-3-1

2 instances selected

Alarm recommendations Investigate with AI - new 3h 1d 1w 1h UTC timezone Explore related

CPU utilization (%)

Percent

23.4%

11.7%

0%

06:00 06:30

i-038c22fc550ffc76

Network in (bytes)

Bytes

36.66M

18.33M

0

06:00 06:30

i-038c22fc550ffc76

Network out (bytes)

Bytes

233.1k

116.6k

0

06:00 06:30

i-038c22fc550ffc76

Network packets in (count)

Count

25.33K

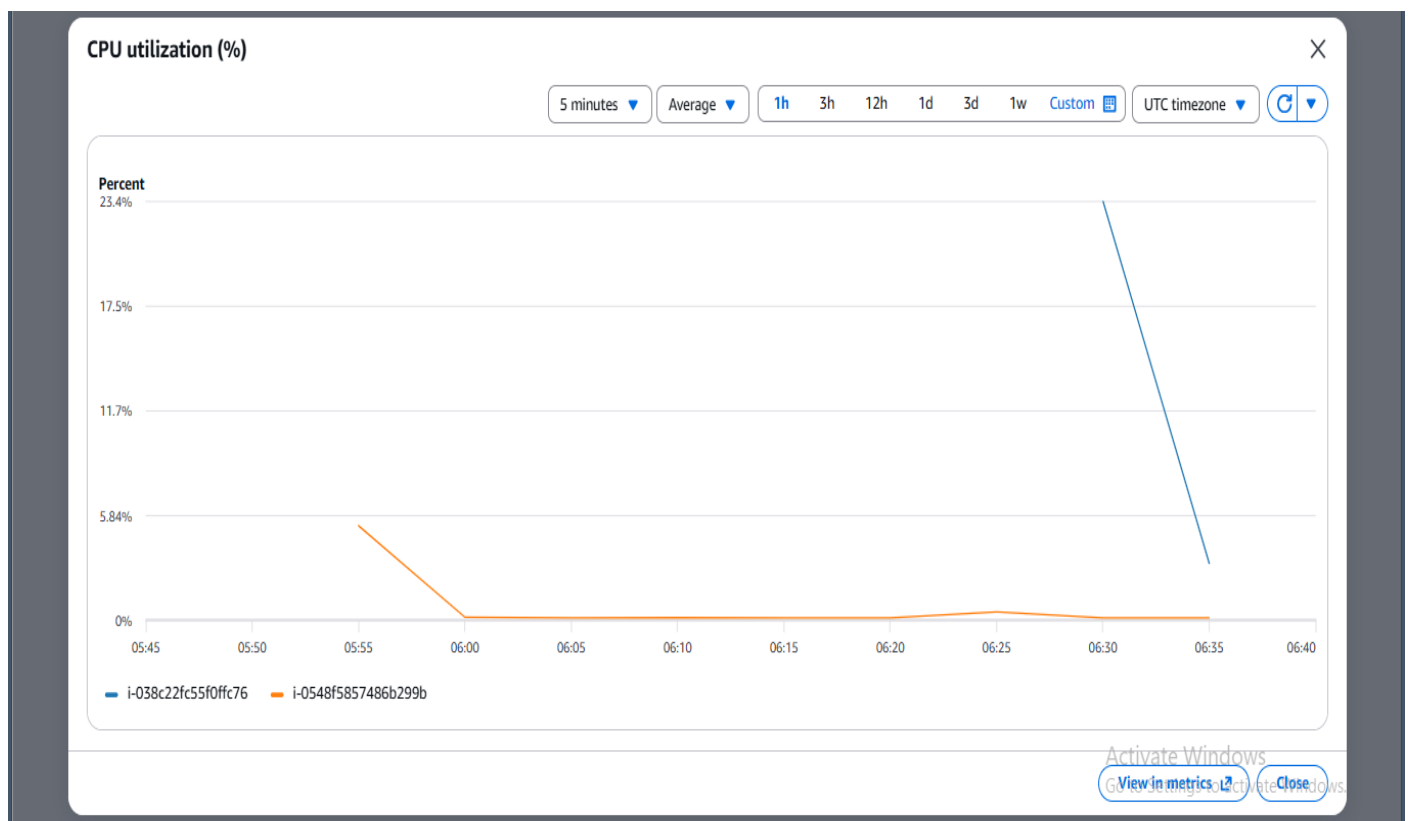
12.66K

0

06:00 06:30

i-038c22fc550ffc76

16. After a while the instance reaches and surpasses it's 50% CPU utilisation threshold.



17. So to take that load off the instance, the auto scaler spins up another instance to balance the load.

The figure is a screenshot of the AWS Management Console showing the EC2 Instances page. The page displays a list of instances, including their names, IDs, states, types, status checks, alarm statuses, availability zones, and public IP addresses. The instances are sorted by name. The table shows three instances: one terminated, one initializing, and one running. The running instance is the one that was spun up by the auto scaler to balance the load.

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IP
	i-0b977c10770124b74	Terminated	t3.micro	-	View alarms +	ap-south-1b	-
	i-038c22fc55f0ffc76	Initializing	t3.micro	Initializing	View alarms +	ap-south-1b	ec2-15-
	i-0548f5857486b299b	Running	t3.micro	3/3 checks passed	View alarms +	ap-south-1a	ec2-3-1